Classification of tumor as Malignant/Non-Malignant using SVM

ML Lab — CA2 Mini Project Report

Machine Learning | Scikit-learn | Python | Wisconsin Breast Cancer Dataset

Group Members

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Dataset: Wisconsin Breast Cancer (sklearn) **Algorithm:** Support Vector Machine

Language: Python 3 **Libraries:** scikit-learn, pandas, matplotlib, joblib

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1 Problem Statement

Breast cancer is one of the most common and dangerous diseases globally. Early and precise detection of tumor type, **malignant** (cancerous) or **benign** (non-cancerous), is key for effective treatment and better survival rates.

Problem: Using 30 numeric features taken from digitized images of fine needle aspirate (FNA) of breast masses, create a machine learning classifier that accurately predicts whether a tumor is **malignant** or **benign**.

The main challenges in this project are:

- High dimensionality (30 features per sample)
- Class imbalance between malignant and benign samples
- Minimising false negatives (missed malignant cases), which is clinically critical
- Choosing the best SVM kernel and hyperparameters

2 Introduction

2.1 Background

Support Vector Machines (SVM) are supervised learning models that work well for binary classification problems with high-dimensional feature spaces. An SVM finds the best **hyperplane** that separates two classes in the feature space, making it suitable for medical diagnosis tasks.

2.2 Dataset Description

The **Wisconsin Breast Cancer Dataset** (from `sklearn.datasets.load_breast_cancer`) contains:

Attribute	Value
Total samples	569
Features	30 numeric (mean, SE, worst of 10 cell nucleus measurements)
Classes	2 — Malignant (212), Benign (357)
Missing values	None

Table 1: Dataset summary

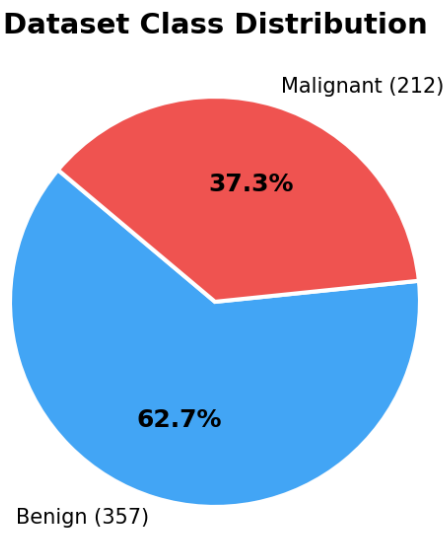

Figure 1: Class distribution in the Wisconsin Breast Cancer dataset

2.3 Why SVM?

- Effective in high-dimensional spaces
- Memory-efficient (uses only support vectors)
- Flexible with the kernel trick (linear, RBF, polynomial)
- Strong track record in medical classification tasks

3 Methodology and Block Diagram

3.1 Pipeline Overview

The end-to-end pipeline has 9 stages as shown in the block diagram below:

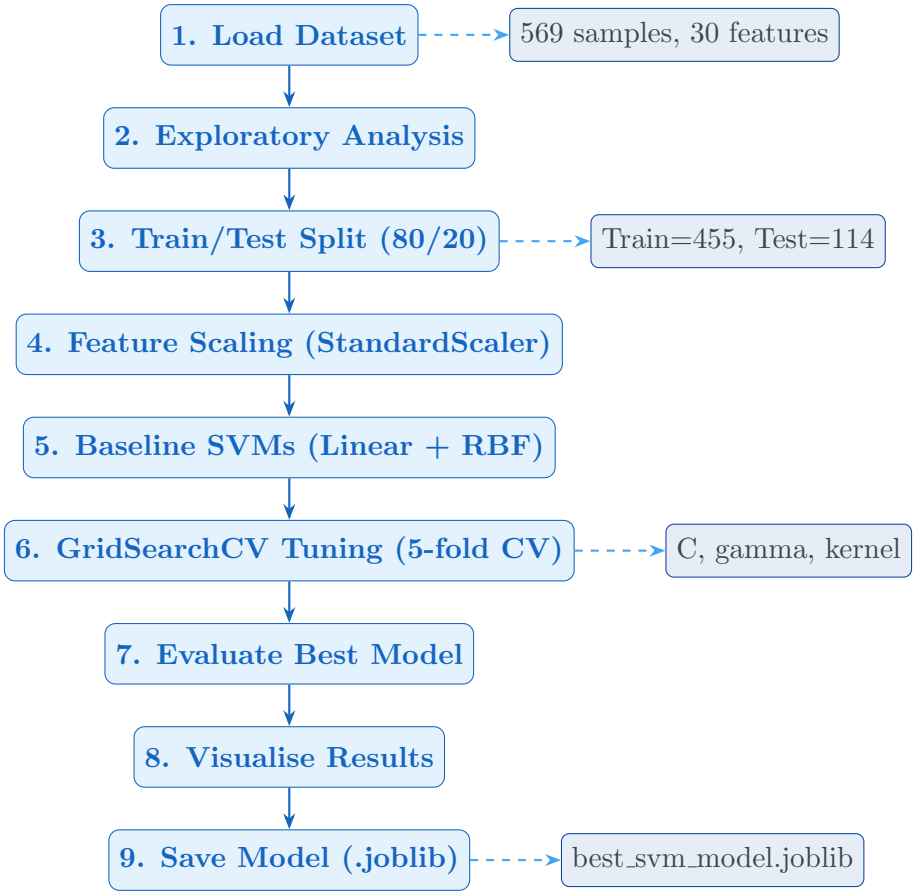

Figure 2: End-to-end SVM classification pipeline

3.2 Hyperparameter Search Space

Parameter	Values Searched	Best Value
`C`	0.1, 1, 10, 100	**10**
`gamma`	0.001, 0.01, 0.1, 1, auto	**0.01**
`kernel`	linear, rbf	**rbf**
Total combinations		40
CV folds		5 (200 fits total)
Scoring metric		F1

Table 2: GridSearchCV hyperparameter space

4 Pseudocode and Flowchart

4.1 Pseudocode

Algorithm 1 SVM Breast Cancer Classifier

Input: Wisconsin Breast Cancer Dataset
Output: Trained SVM model, evaluation metrics

$X, y \leftarrow$ `load_breast_cancer()`
$X_{train}, X_{test}, y_{train}, y_{test} \leftarrow$ `train_test_split`$(X, y, 0.8)$
$scaler \leftarrow$ `StandardScaler().fit`(X_{train})
$X^s_{train}, X^s_{test} \leftarrow scaler$.transform$(X_{train}, X_{test})$

for kernel $\in$ {linear, rbf} **do**
    svm $\leftarrow$ `SVC(kernel=kernel).fit`(X^s_{train}, y_{train})
    Print accuracy on X^s_{test}
end for

pipeline $\leftarrow$ `Pipeline([StandardScaler, SVC])`
param_grid $\leftarrow \{C : [0.1, 1, 10, 100],\ \gamma : [0.001, 0.01, 0.1, 1],\ kernel : [\text{linear, rbf}]\}$
gs $\leftarrow$ `GridSearchCV(pipeline, param_grid, cv=5, scoring=f1)`
gs.fit(X_{train}, y_{train})
best_model $\leftarrow$ gs.best_estimator_

$\hat{y} \leftarrow$ best_model.predict(X_{test})
Compute: accuracy, precision, recall, F1, confusion matrix
Plot confusion matrix heatmap
`joblib.dump`(best_model, `"best_svm_breast_cancer_model.joblib"`)

4.2 Flowchart

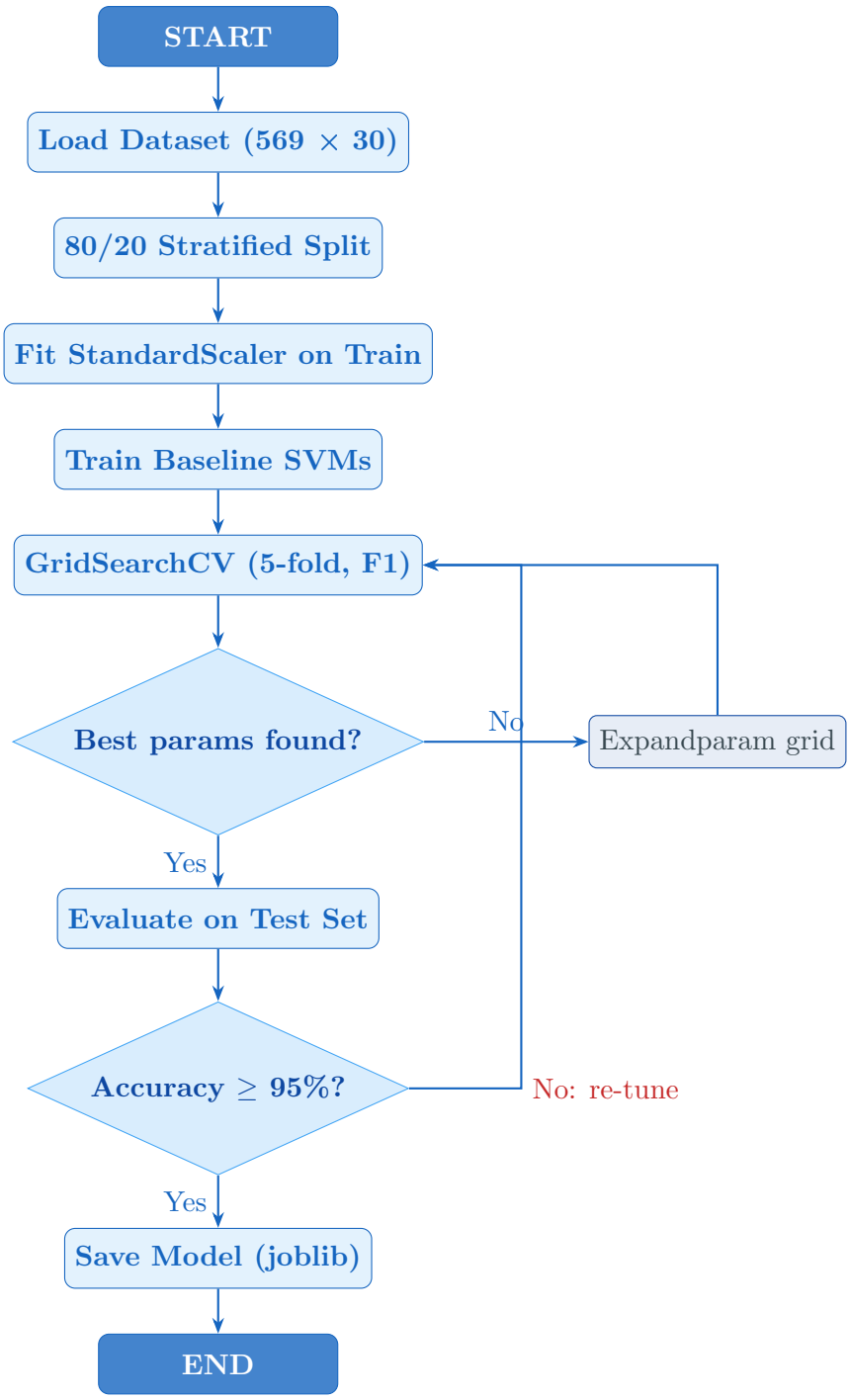

Figure 3: Detailed flowchart of the classification system

5 Results and Analysis

5.1 Baseline Model Comparison

Model	Kernel	Test Accuracy
Baseline SVM 1	Linear	0.9737 (97.37%)
Baseline SVM 2	RBF	0.9825 (98.25%)
Tuned SVM (Best)	**RBF**	**0.9825 (98.25%)**

Table 3: Kernel accuracy comparison

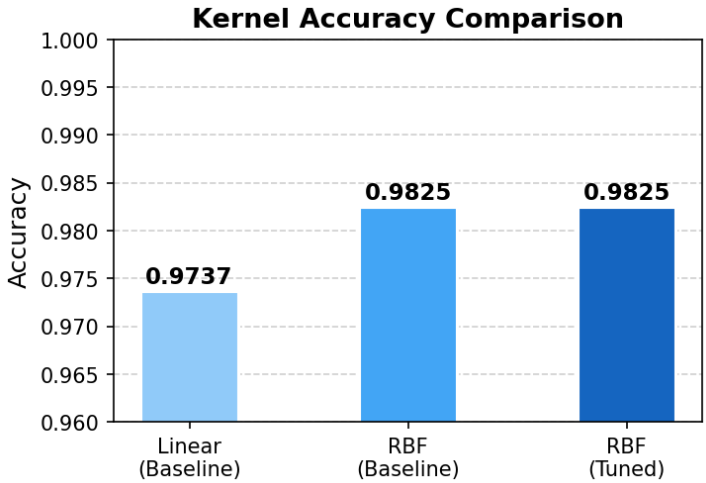

Figure 4: Test accuracy across kernel configurations

5.2 Best Model Parameters

`kernel = rbf` | `C = 10` | `gamma = 0.01`

Best CV F1 Score: 0.9844

5.3 Classification Report

Class	Precision	Recall	F1-Score	Support
Malignant (0)	0.98	0.98	0.98	42
Benign (1)	0.99	0.99	0.99	72
Accuracy		**0.9825**		**114**
Macro avg	0.98	0.98	0.98	114
Weighted avg	0.98	0.98	0.98	114

Table 4: Full classification report on the held-out test set

5.4 Performance Metrics Visualisation

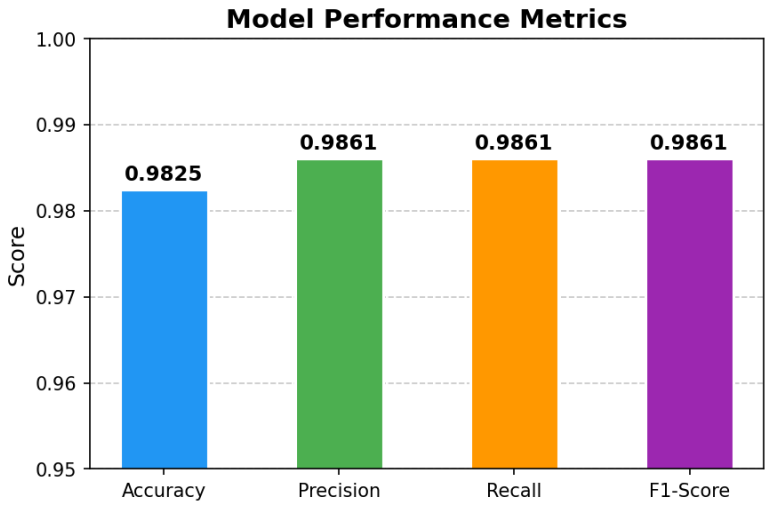

Figure 5: Accuracy, Precision, Recall and F1-Score of the best model

5.5 Confusion Matrix

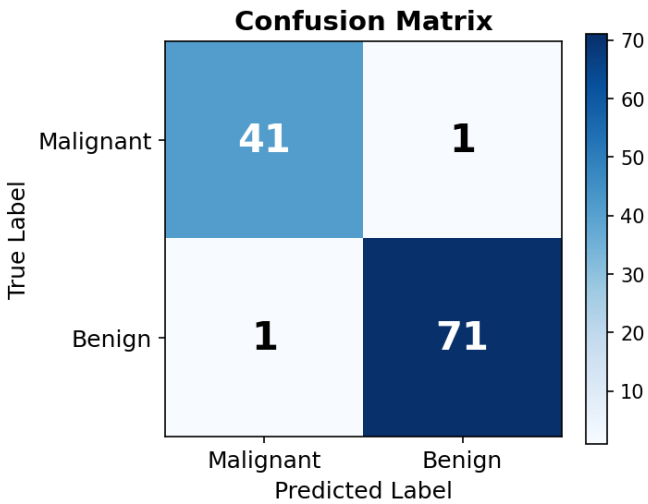

Figure 6: Confusion matrix on the 114-sample test set

From the confusion matrix, we can observe:

- **True Positives (Malignant correctly predicted):** 41
- **True Negatives (Benign correctly predicted):** 71
- **False Positives (Benign predicted as Malignant):** 1
- **False Negatives (Malignant missed):** 1

Clinical significance: Only **1 false negative** out of 42 malignant cases means the model missed only one actual cancer. This results in a **recall of 97.6%** for malignant tumors, indicating excellent performance.

6 Conclusion

This project successfully developed an end-to-end SVM classifier for breast cancer tumor detection using the Wisconsin Breast Cancer dataset. Key findings include:

- The **RBF kernel with C=10 and gamma=0.01** provided the best performance after tuning.
- The model achieved **98.25% accuracy** and **98.61% F1-score** on the held-out test set.
- Only **2 misclassifications out of 114 samples** (1 FP and 1 FN), which shows strong generalisation.
- Feature scaling using **StandardScaler** was crucial and applied within the CV pipeline to avoid data leakage.
- SVM proved to be a strong, interpretable choice for high-dimensional medical classification tasks.

Future scope:

- Wrap model as a REST API (FastAPI or Flask) for clinical use
- Add LIME-based feature importance for explainability
- Compare with ensemble methods (Random Forest, XGBoost)
- Create a Jupyter Notebook version for interactive exploration

7 References

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